VIDYA NGO • TECHFEST 2026

Logo / Team Photo

SMART FLOOD-ALERT BRIDGE

A water-level sensing bridge that rises automatically, sounds an alarm, and lights up to keep communities safe during floods.

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INTRODUCTION

Why This Project?

As part of VIDYA NGO's Techfest, our team was asked to identify a real problem faced by our community and design a working solution using simple, affordable technology. We looked around our own locality and found a challenge that affects hundreds of people every monsoon.

Our Goal

Build a low-cost smart system that warns and protects people at flood-prone bridges in real time.

Locality / Site Photo

THE PROBLEM

Rising River Water Cuts Off the Bridge

During heavy rainfall, the river water level rises quickly and floods the bridge connecting our area. This is a recurring problem every monsoon season.

> *“Every time it rains heavily, the bridge goes under water — and our locality gets completely cut off from the rest of the city.”*

Flooded Bridge Photo

THE PROBLEM

What Exactly Goes Wrong

Heavy rainfall causes floods

Monsoon rain makes the river rise faster than people can react.

Bridge becomes impassable

Rising water submerges the bridge, blocking the only route across.

People stranded for hours or days

Commuters, students and residents are left waiting with no safe way to cross.

Emergency services can't respond

Ambulances and rescue vehicles are delayed exactly when speed matters most.

THE PROBLEM

Who Suffers Because of This

Students

Miss school days and exams when the bridge floods on their route.

Commuters & Vehicles

Cars and two-wheelers are stuck for hours with no alternate path.

Emergency Services

Ambulances, fire and rescue teams face life-threatening delays.

Local Residents

Families are cut off from markets, hospitals and workplaces for days.

Result: repeated disruption, safety risk, and economic loss for the whole community every rainy season.

OUR SOLUTION

How It Works

1
Water Rises

Water level sensor is placed on the river near the bridge.

2
Arduino Reads Signal

Arduino UNO continuously monitors the sensor's readings.

3
Threshold Crossed

When water crosses the safe limit, Arduino triggers the alert.

4
Buzzer + LED Activate

The buzzer beeps and LED glows to warn nearby people.

5
Bridge Lifts Up

The bridge deck rises, keeping the path safe and passable.

OUR SOLUTION

Components We Used

Arduino UNO

The brain of the project — reads sensor data and controls outputs.

Water Level Sensor

Detects how high the river water has risen.

Alarm Buzzer

Beeps loudly to alert people nearby of rising danger.

9V Battery

Powers the Arduino and connected components.

Breadboard

Connects all components together without soldering.

LED

Glows as a visual warning signal alongside the buzzer.

OUR SOLUTION

Our Working Model

Photos and circuit diagram of the prototype will be added here.

Circuit Diagram

Working Model Photo

BENEFITS

Why This Solution Helps

Early Warning

Alerts people before the bridge becomes dangerous to cross.

Prevents Accidents

Reduces the risk of vehicles or pedestrians entering flood water.

Saves Time

People can plan alternate routes instead of waiting blindly.

Faster Emergency Response

Keeps a safe path open for ambulances and rescue teams.

Low Cost

Built using affordable, easily available components.

Simple to Maintain

Easy to install, operate, and repair with basic tools.

FUTURE SCOPE

Where This Project Can Go

Mobile App Alerts

Send real-time flood warnings directly to residents' phones.

IoT Cloud Connectivity

Connect sensors to the cloud so authorities can monitor many bridges at once.

Solar-Powered Unit

Replace battery with solar panels for uninterrupted, eco-friendly operation.

Scale to Other Locations

Deploy the same system on other flood-prone bridges and low-lying roads.

Link to Emergency Services

Automatically notify local disaster management and rescue teams.

Traffic Signal Integration

Automatically switch nearby traffic signals to guide vehicles to safe routes.

Together, We Can Keep Our Community Safe

A small, affordable sensor system can prevent stranded students, blocked ambulances, and flood-related suffering — turning a yearly crisis into a manageable, well-warned event.

THANK YOU

Anjum Shaikh | VIDYA NGO Techfest | Smart Flood-Alert Bridge